

No. of Functional Features Included in the Solution

- OptiCrop

S.No	Feature	Description	Module	Status	Marks
1	Crop Recommendation	Predicts suitable crop	ML Model	Done	1
2	Soil Input	Accepts soil/environment data	Frontend	Done	0.5
3	Prediction Engine	Processes user input	Backend	Done	1
4	Web Interface	Displays recommendations	Flask UI	Done	0.5
5	Dataset Processing	Prepares training data	Data	Done	0.5
6	Model Training	Trains ML model	ML	Done	0.5
7	GitHub Repository	Version control	Repository	Done	0.5
8	Future Weather API	Live weather integration	API	Pending	0.5

Metric	Value
Total Features Planned	8
Total Features Implemented	7
Core/Must-Have Features	Crop Recommendation, Prediction Engine, Web Interface
Additional/Nice-to-Have	Weather API, Fertilizer Recommendation
Features Tested & Verified	Input validation, Prediction, UI, Model loading

Feature Category Breakdown

S.No	Category	Features	Examples
1	User Interface	Input forms, Result page	Flask HTML templates
2	Backend/Logic	Prediction logic	Flask + Python
3	Database/Storage	Dataset & Model	CSV, PKL
4	API/Integration	Future Weather API	OpenWeather (planned)
5	Security	Input validation	Basic validation